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Biotechnology and It's Applications

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MCQ (Single Correct Answer)	97 questions
Answer key	at the end

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MCQ (Single Correct Answer) 97 questions

Q1. RE-NEET 2026

The inactive form of Bt toxin is converted to the active form in the insect gut ____

(A) by nucleases
(B) due to alkaline pH
(C) due to acidic pH
(D) by proteases

Open solution

Q2. NEET 2026

The human protein named α-1-antitrypsin, obtained from transgenic animals, is used for the treatment of ____ .

(A) Emphysema
(B) Alzheimer's disease
(C) Rheumatoid arthritis
(D) Cystic fibrosis

Open solution

Q3. NEET 2026

The toxin proteins isolated from Bacillus thuringiensis, coded by which of the following genes would control cotton bollworms and corn borer, respectively?

(A) cryIAc and cryIAb
(B) cryIAc and cryIIAb
(C) cryIIAb and cryIAc
(D) cryIAc and cryIIIAb

Open solution

Q4. NEET 2025

Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin?

(A) Virus (B) Phage (C) Bacterium (D) Yeast

Open solution

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Q5. NEET 2024 (Re-Examination)

Open solution

Following are the steps involved in action of toxin in Bt. Cotton

A. The inactive toxin converted into active form due to alkaline pH of gut of insect.

B. Bacillus thuringiensis produces crystals with toxic insecticidal proteins.

C. The alkaline pH solubilises the crystals.

D. The activated toxin binds to the surface of midgut cells, creates pores and causes death of the insect.

E. The toxin proteins exist as inactive protoxins in bacteria.

Choose the correct sequence of steps from the options given below:

(A) $\mathrm{E} \rightarrow \mathrm{C} \rightarrow \mathrm{B} \rightarrow \mathrm{A} \rightarrow \mathrm{D}$

(B) $\mathrm{B} \rightarrow \mathrm{C} \rightarrow \mathrm{A} \rightarrow \mathrm{E} \rightarrow \mathrm{D}$

(C) $\mathrm{A} \rightarrow \mathrm{E} \rightarrow \mathrm{B} \rightarrow \mathrm{D} \rightarrow \mathrm{C}$

(D) $\mathrm{B} \rightarrow \mathrm{E} \rightarrow \mathrm{C} \rightarrow \mathrm{A} \rightarrow \mathrm{D}$

Q6. NEET 2024

Open solution

The capacity to generate a whole plant from any cell of the plant is called:

(A) Totipotency

(B) Micropropagation

(C) Differentiation

(D) Somatic hybridization

Q7. NEET 2024

Open solution

Given below are two statements:

Statement I : Bt toxins are insect group specific and coded by a gene cry IAc.

Statement II : Bt toxin exists as inactive protoxin in B. thuringiensis. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.

In the light of the above statements, choose the correct answer from the options given below:

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

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Q8. NEET 2024

What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the genome of the recipient.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of chromosome.

E. It shows ability to replicate.

Choose the correct answer from the options given below:

(A) A and B only **(B)** D and E only **(C)** B and C only **(D)** A and E only

Open solution

Q9. NEET 2024

Which of the following are fused in somatic hybridization involving two varieties of plants?

(A) Callus **(B)** Somatic embryos **(C)** Protoplasts **(D)** Pollens

Open solution

Q10. NEET 2024

Match List I with List II

	List - I		List - II
A.	α -I antitrypsin	I.	Cotton bollworm
B.	Cry IAb	II.	ADA deficiency
C.	Cry IAc	III.	Emphysema
D.	Enzyme replacement therapy	IV.	Corn borer

Choose the correct answer form the options given below:

(A) A-II, B-I, C-IV, D-III **(B)** A-III, B-I, C-II, D-IV

(C) A-III, B-IV, C-I, D-II **(D)** A-II, B-IV, C-I, D-III

Open solution

Q11. NEET 2024

Which of the following statements is incorrect?

(A) A bio-reactor provides optimal growth conditions for achieving the desired product

(B) Most commonly used bio-reactors are of stirring type

(C) Bio-reactors are used to produce small scale bacterial cultures

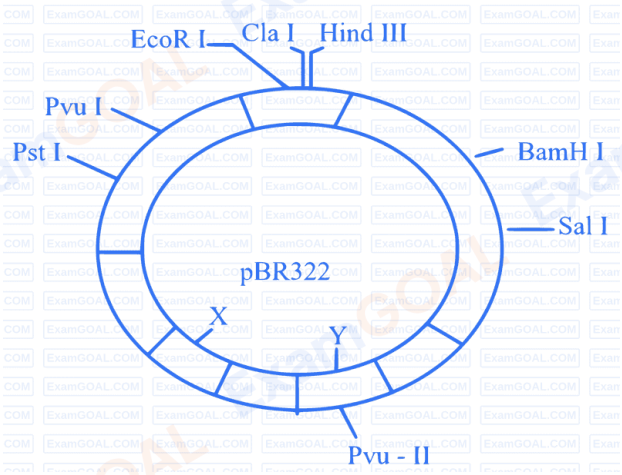
(D) Bio-reactors have an agitator system, an oxygen delivery system and foam control system

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Q12. NEET 2024

The following diagram shows restriction sites in E. coli cloning vector pBR322. Find the role of 'X' and 'Y' genes :



Open solution

(A) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of Plasmid.

(B) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of Plasmid.

(C) The gene 'X' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.

(D) Gene 'X' is responsible for recognition sites and 'Y' is responsible for antibiotic resistance.

Q13. NEET 2023 Manipur

Which of the following statement is incorrect about Agrobacterium tumifaciens?

Open solution

(A) It is used to deliver gene of interest in both prokaryotic as well as eukaryotic host cells.

(B) 'Ti' plasmid from Agrobacterium tumifaciens used for gene transfer is not pathogenic to plant cell.

(C) It transforms normal plant cells into tumor cells.

(D) It delivers 'T-DNA' into plant cell.

Q14. NEET 2023 Manipur

Which of the following can act as molecular scissors?

Open solution

(A) Restriction enzymes

(B) DNA ligase

(C) RNA polymerase

(D) DNA polymerase

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Q15. NEET 2023 Manipur

Match List-I with List-II.

	List - I		List - II
(A)	Gene therapy	(I)	Separation of DNA fragments
(B)	RNA interference	(II)	Diagnostic test for AIDS
(C)	ELISA	(III)	Cellular defence
(D)	Gel Electrophoresis	(IV)	Allows correction of a gene defect.

Choose the correct answer from the options given below :

(A) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)
(B) (A)-(IV), (B)-(II), (C)-(III), (D)-(I)
(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(D) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

Open solution

Q16. NEET 2023

During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

(A) DNA **(B)** Histones **(C)** Polysaccharides **(D)** RNA

Open solution

Q17. NEET 2023

In gene gun method used to introduce alien DNA into host cells, microparticles of _____ metal are used.

(A) Zinc **(B)** Tungsten or gold **(C)** Silver **(D)** Copper

Open solution

Q18. NEET 2023

Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

A. Insertion of recombinant DNA into the host cell

B. Cutting of DNA at specific location by restriction enzyme

C. Isolation of desired DNA fragment

D. Amplification of gene of interest using PCR

Choose the correct answer from the options given below :

(A) C, A, B, D **(B)** C, B, D, A **(C)** B, D, A, C **(D)** B, C, D, A

Open solution

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Q19. NEET 2022 Phase 2

Milk of transgenic 'Cow Rosie' was nutritionally more balanced product for human babies than natural cow milk because it contained:

Open solution

(A) Human enzyme Adenosine Deaminase (ADA)

(B) Human protein α-1-antitrypsin

(C) Human alpha-lactalbumin

(D) Human insulin-like growth factor

Q20. NEET 2022 Phase 1

In the following palindromic base sequences of DNA, which one can be cut easily by particular restriction enzyme?

Open solution

(A) 5'GATACT3' ; 3'CTATGA5'

(B) 5'GAATTC3' ; 3'CTTAAG5'

(C) 5'CTCAGT3' ; 3'GAGTCA5'

(D) 5'GTATTC3' ; 3'CATAAG5'

Q21. NEET 2022 Phase 1

In gene therapy of Adenosine Deaminase (ADA) deficiency, the patient requires periodic infusion of genetically engineered lymphocytes because:

Open solution

(A) Retroviral vector is introduced into these lymphocytes.

(B) Gene isolated from marrow cells producing ADA is introduced into cells at embryonic stages

(C) Lymphocytes from patient's blood are grown in culture, outside the body

(D) Genetically engineered lymphocytes are not immortal cells.

Q22. NEET 2022 Phase 1

Statements related to human Insulin are given below.

Which statements is/are correct about genetically engineered Insulin?

Open solution

(a) Pro-hormone insulin contain extra stretch of C-peptide

(b) A-peptide and B-peptide chains of insulin were produced separately in E.coli, extracted and combined by creating disulphide bond between them.

(c) Insulin used for treating Diabetes was extracted from Cattles and Pigs.

(d) Pro-hormone Insulin needs to be processed for converting into a mature and functional hormone.

(e) Some patients develop allergic reactions to the foreign insulin.

Choose the most appropriate answer from the options given below :

(A) (a), (b) and (d) only

(B) (b) only

(C) (c) and (d) only

(D) (c), (d) and (e) only

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Q23. NEET 2021

When gene targetting involving gene amplification is attempted in an individual's tissue to treat disease, it is known as :

(A) Safety testing
(B) Biopiracy
(C) Gene therapy
(D) Molecular diagnosis

Open solution

Q24. NEET 2021

Now-a-days it is possible to detect the mutated gene causing cancer by allowing radioactive probe to hybridise to its complementary DNA in a clone of cells, followed by its detection using autoradiography because:

(A) mutated gene does not appear on photographic film as the probe has complementarity with it
(B) mutated gene partially appears on a photographic film
(C) mutated gene completely and clearly appears on a photographic film
(D) mutated gene does not appear on a photographic film as the probe has no complementarity with it

Open solution

Q25. NEET 2021

With regard to insulin choose correct options.

(1) C-peptide is not present in mature insulin.

(2) The insulin produced by rDNA technology has C-peptide.

(3) The pro-insulin has C-peptide

(4) A-peptide and B-peptide of insulin are interconnected by disulphide bridges.

Choose the correct answer from the options given below.

(A) (1) and (4) only
(B) (2) and (4) only
(C) (2) and (3) only
(D) (1), (3) and (4) only

Open solution

Q26. NEET 2020 Phase 2

The laws and rules to prevent unauthorised exploitation of bio-resources are termed as -

(A) Biopiracy
(B) Biopatenting
(C) Bioethics
(D) Bioengineering

Open solution

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Q27. NEET 2020 Phase 2

RNA interference is used for which of the following purposes in the field of biotechnology ?

(A) to reduce post harvest losses

(B) to develop a plant tolerant to abiotic stresses

(C) to develop a pest resistant plant against infestation by nematode

(D) to enhance the mineral usage by the plant

Open solution

Q28. NEET 2020 Phase 1

Bt cotton variety that was developed by the introduction of toxin gene of Bacillus thuringiensis (Bt) is resistant to

(A) Insect predators (B) Insect pests (C) Fungal diseases (D) Plant nematodes

Open solution

Q29. NEET 2020 Phase 1

Which of the following statements is not correct?

(A) In man insulin is synthesised as a proinsulin

(B) The proinsulin has an extra peptide called C-peptide.

(C) Genetically engineered insulin is produced in E.Coli.

(D) The functional insulin has A and B chains linked together by hydrogen bonds.

Open solution

Q30. NEET 2020 Phase 1

Match the following columns and select the correct option.

Column-I	Column-II
(a) Bt cotton	(i) Gene therapy
(b) Adenosine deaminase deficiency	(ii) Cellular defence
(c) RNAi	(iii) Detection of HIV infection
(d) PCR	(iv) Bacillus thuringiensis

(A) (A) $\rightarrow$ (i), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (iii), (D) $\rightarrow$ (iv)

(B) (A) $\rightarrow$ (iii), (B) $\rightarrow$ (ii), (C) $\rightarrow$ (i), (D) $\rightarrow$ (iv)

(C) (A) $\rightarrow$ (iv), (B) $\rightarrow$ (i), (C) $\rightarrow$ (ii), (D) $\rightarrow$ (iii)

(D) (A) $\rightarrow$ (ii), (B) $\rightarrow$ (iii), (C) $\rightarrow$ (iv), (D) $\rightarrow$ (i)

Open solution

Q31. NEET 2019 (Odisha)

In RNAi, the genes are silenced using :

(A) ds - DNA (B) ds - RNA (C) ss - DNA (D) ss - RNA

Open solution

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Q32. NEET 2019 (Odisha)

Exploitation of bioresources of a nation by multinational companies without authorization from the concerned country is referred to as

(A) Biowar (B) Bioweapon (C) Biopiracy (D) Bioethics

Open solution

Q33. NEET 2019

Which of the following glucose transporters is insulin-dependent?

(A) GLUT III (B) GLUT IV (C) GLUT I (D) GLUT II

Open solution

Q34. NEET 2018

Which of the following is commonly used as a vector for introducing a DNA fragment in human lymphocytes?

(A) Retrovirus (B) Ti plasmid (C) λ phage (D) pBR322

Open solution

Q35. NEET 2018

In India, the organisation responsible for assessing the safety of introducing genetically modified organisms for public use is

(A) Indian Council of Medical Research (ICMR)

(B) Council for Scientific and Industrial Research (CSIR)

(C) Research Committee on Genetic Manipulation (RCGM)

(D) Genetic Engineering Appraisal Committee (GEAC)

Open solution

Q36. NEET 2018

A 'new' variety of rice was patented by a foreign company, though such varieties have been present in India for a long time. This is related to

(A) Co-667 (B) Sharbati Sonora (C) Lerma Rojo (D) Basmati

Open solution

Q37. NEET 2018

Use of bioresources by multinational companies and organisations without authorisation from the concerned country and its people is called

(A) bio-infringement (B) biodegradation (C) biopiracy (D) bioexploitation

Open solution

Q38. NEET 2016 Phase 2

Which kind of therapy was given in 1990 to a four-year-old girl with adenosine deaminase [ADA] deficiency ?

(A) Radiation therapy (B) Immunotherapy

(C) Gene therapy (D) Chemotherapy

Open solution

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Q39. NEET 2016 Phase 1

The two polypeptides of human insulin are linked together by

(A) Hydrogen bonds
(B) Covalent bond
(C) Disulphide bridges
(D) Phosphodiester bond

Open solution

Q40. NEET 2016 Phase 1

Which part of the tobacco plant is infected by Meloidogyne incognita ?

(A) Flower
(B) Stem
(C) Leaf
(D) Root

Open solution

Q41. AIPMT 2015

Golden rice is a genetically modified crop plant where the incorporated gene is meant for biosynthesis of -

(A) Vitamin C
(B) Omega 3
(C) Vitamin A
(D) Vitamin B

Open solution

Q42. AIPMT 2015

The introduction of t-DNA into plants involves -

(A) Altering the pH of the soil, then heat-shocking the plants
(B) Infection of the plant by Agrobacterium tumefaciens
(C) Exposing the plants to cold for a brief period
(D) Allowing the plant roots to stand in water

Open solution

Q43. AIPMT 2015 Cancelled Paper

In Bt cotton, the Bt toxin present in plant tissue as pro - toxin is converted into active toxin due to :

(A) action of gut micro-organisms
(B) presence of conversion factors in insect gut
(C) alkaline pH of the insect gut
(D) acidic pH of the insect gut

Open solution

Q44. AIPMT 2015 Cancelled Paper

Which body of the Government of India regulates GM research and safety of introducing GM organisms for public services ?

(A) Research Committee on Genetic Manipulation
(B) Genetic Engineering Approval Committee
(C) Indian Council of Agricultural Research
(D) Bio - safety committee

Open solution

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Q45. AIPMT 2015 Cancelled Paper

The crops engineered for glyphosate are resistant/ tolerant to :

(A) Bacteria (B) Herbicides (C) Fungi (D) Insects

Open solution

Q46. AIPMT 2014

To obtain virus−free healthy plants from a diseased one by tissue culture technique, which part/parts of the diseased plant will be taken ?

(A) Epidermis only (B) Palisade parenchyma

(C) Both apical and axillary meristems (D) Apical meristem only

Open solution

Q47. AIPMT 2014

The first human hormone produced by recombinant DNA technology is :

(A) Estrogen (B) Insulin (C) Thyroxin (D) Progesterone

Open solution

Q48. NEET 2013 (Karnataka)

Tissue culture technique can produce infinite number of new plants from a small parental tissue. The economic importance of the technique is in raising:

(A) development of new species

(B) genetically uniform population identical to the original parent

(C) variants through picking up somaclonal variations

(D) homozygous diploid plants

Open solution

Q49. NEET 2013 (Karnataka)

Which of the following statements is not true about somatic embryogenesis ?

(A) Somatic embryos can develop from microspores

(B) Somatic embryo is induced usually by an auxin such as 2, 4−D

(C) The pattern of development of a somatic embryo is comparable to that of a zygotic embryo

(D) A somatic embryo develops from a somatic cell

Open solution

Q50. NEET 2013 (Karnataka)

RNA interference involves :-

(A) Interference of RNA in synthesis of DNA

(B) Synthesis of cDNA from RNA using reverse transcriptase

(C) Synthesis of mRNA from DNA

(D) Silencing of specific mRNA due to complementary RNA

Open solution

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Q51. NEET 2013 (Karnataka)

Which one of the following vectors is used to replace the defective gene in gene therapy ?

(A) Ti plasmid (B) Cosmid (C) Ri plasmid (D) Adenovirus

Open solution

Q52. NEET 2013

Which of the following Bt crops is being grown in India by the farmers ?

(A) Soybean (B) Cotton (C) Brinjal (D) Maize

Open solution

Q53. AIPMT 2012 Mains

Tobacco plants resistant to a nematode have been developed by the introduction of DNA that produced (in the host cells) :

(A) An antifeedant (B) A toxic protein

(C) Both sense and anti-sense RNA (D) A particular hormone

Open solution

Q54. AIPMT 2012 Mains

What is it that forms the basis of DNA Fingerprinting ?

(A) The relative proportions of purines and pyrimidines in DNA

(B) The relative amount of DNA in the ridges and grooves of the fingerprints

(C) The relative difference in the DNA occurrence in blood, skin and saliva

(D) Satellite DNA occurring as highly repeated short DNA segments

Open solution

Q55. AIPMT 2012 Mains

The first clinical gene therapy was given for treating -

(A) Rheumatoid arthritis (B) Adenosine deaminase deficiency

(C) Diabetes mellitus (D) Chicken pox

Open solution

Q56. AIPMT 2012 Prelims

Which one of the following is a case of wrong matching ?

(A) Somatic hybridization - Fusion of two diverse cells

(B) Callus - Unorganised mass of cells produced in tissue culture

(C) Vector DNA - site for tRNA synthesis

(D) Micropropagation - In vitro production of plants in large numbers

Open solution

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Q57. AIPMT 2012 Prelims

Which part would be most suitable for raising virus free plants for microporpagation ?

(A) Vascular tissue (B) Node (C) Bark (D) Meristem

Open solution

Q58. AIPMT 2012 Prelims

Consumption of which one of the following foods can prevent the kind of blindness associated with vitamin 'A' deficiency

(A) Flaver Savr Tomato (B) Canolla

(C) Bt- Brinjal (D) Golden rice

Open solution

Q59. AIPMT 2011 Mains

Bacillus thuringiensis forms protein crystals which contain insecticidal protein. This protein :

(A) is activated by acid pH of the foregut of the insect pest.

(B) binds with epithelial cells of midgut of the insect pest ultimately killing it

(C) is coded by several genes including the gene cry

(D) does not kill the carrier bacterium which is itself resistant to this toxin

Open solution

Q60. AIPMT 2011 Mains

Silencing of mRNA has been used in producing transgenic plants resistant to :

(A) White rusts (B) Nematodes

(C) Bacterial blights (D) Bollworms

Open solution

Q61. AIPMT 2011 Mains

Read the following four statements (A-D) about certain mistakes in two of them :
(A) The first transgenic buffalo, Rosie produced milk which was human alpha-lactalbumin enriched.
(B) Restriction enzymes are used in isolation of DNA from other macro-molecules
(C) Downstream processing is one of the steps of R-DNA technology
(D) Disarmed pathogen vectors are also used in transfer of R-DNA into the host
Which are the two statements having mistakes ?

(A) Statements (B) and (C) (B) Statements (A) and (C)

(C) Statements (A) and (B) (D) Statements (C) and (D)

Open solution

Q62. AIPMT 2011 Prelims

Maximum number of existing transgenic animals is of :

(A) Mice (B) Cow (C) Pig (D) Fish

Open solution

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Q63. AIPMT 2011 Prelims

The process of RNA interference has been used in the development of plants resistant to :

(A) Nematodes (B) Fungi (C) Viruses (D) Insects

Open solution

Q64. AIPMT 2010 Mains

Which one of the following is now being commercially produced by biotechnological procedures ?

(A) Insulin (B) Morphine (C) Nicotine (D) Quinine

Open solution

Q65. AIPMT 2010 Prelims

Breeding of crops with high levels of minerals, vitamins and proteins is called –

(A) Biofortification (B) Biomagnification

(C) Micorpropagation (D) Somatic hybridisation

Open solution

Q66. AIPMT 2010 Prelims

The genetically-modified (GM) brinjal in India has been developed for:

(A) Enhancing shelf life (B) Insect-resistance

(C) Drought-resistance (D) Enhancing mineral content

Open solution

Q67. AIPMT 2010 Prelims

Genetic engineering has been successfully used for producing –

(A) transgenic models for studying new treatments for certain cardiac disease

(B) Animals like bulls for farm work as they have super power

(C) transgenic mice for testing safety of polio vaccine before use in humans

(D) transgenic Cow-Rosie which produces high fat milk for making ghee

Open solution

Q68. AIPMT 2010 Prelims

Some of the characteristics of Bt cotton are –

(A) High yield and production of toxic protein crystals which kill dipteran pests

(B) High yield and resistance to bollworms

(C) Long fibre and resistance to aphids

(D) Medium yield, long fibre and resistance to beetle pests

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Q69. AIPMT 2010 Prelims

An improved variety of transgenic basmati rice

(A) gives high yield but has no characteristic aroma

(B) Does not require chemical fertilizers and growth hormones

(C) is completely resistant to all insect pests and disease of paddy

(D) gives high yield and is rich in vitamin A

Open solution

Q70. AIPMT 2009

Polyethylene glycol method is used for :

(A) Biodiesel production

(B) Seedless fruit production

(C) Energy production from sewage

(D) Gene transfer without a vector

Open solution

Q71. AIPMT 2009

Somaclones are obtained by : -

(A) Tissue culture

(B) Plant breeding

(C) Genetic engineering

(D) Irradiation Genetic engineering

Open solution

Q72. AIPMT 2009

What is true about Bt toxin ?

(A) The activated toxin enters the ovaries of the pest to sterilise it and thus prevent its multiplication.

(B) Bt protein exists as active toxin in the Bacillus

(C) The concerned Bacillus has antitoxins

(D) The inactive protoxin gets converted into active form in the insect gut

Open solution

Q73. AIPMT 2009

Transgenic plants are the ones -

(A) Produced after protoplast fusion in artifical medium

(B) Produced by a somatic embryo in artificial medium

(C) Generated by introducing foreign DNA in to a cell and regenerating a plant from that cell

(D) Grown in artificial medium after hybridization in the field

Open solution

Q74. AIPMT 2009

The bacterium Bacillus thuringiensis is widely used in contemporary biology as -

(A) Indicator of water pollution

(B) Agent for production of dairy products

(C) Source of industrial enzyme

(D) Insecticide

Open solution

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Q75. AIPMT 2008

What is antisense technology?

(A) RNA polymerase producing DNA

(B) Production of somaclonal variants in tissue cultures

(C) When a piece of RNA that is complementary in sequence is used to stop expression of a specific gene

(D) A cell displaying a foreign antigen used for synthesis of antigens

Open solution

Q76. AIPMT 2008

Cry I endotoxins obtained from Bacillus thuringiensis are effective against:

(A) Mosquitoes (B) Flies (C) Nematodes (D) Boll worms

Open solution

Q77. AIPMT 2008

Main objective of production/use of herbicide resistant GM crops is to:

(A) eliminate weeds from the field without the use of manual labour

(B) eliminate weeds from the field without the use of herbicides

(C) encourage eco-friendly herbicides

(D) reduce herbicide accumulation in food articles for health safety

Open solution

Q78. AIPMT 2008

Human insulin is being commercially produced from a transgenic species of:

(A) Escherichia (B) Mycobacterium (C) Rhizobium (D) Saccharomyces

Open solution

Q79. AIPMT 2008

A transgenic food crop which may help in solving the problem of night blindness in developing countries is

(A) Flavr Savr tomatoes (B) Bt soybean

(C) Starlink maize. (D) Golden rice

Open solution

Q80. AIPMT 2007

A genetically engineered micro-organism used successfully in bioremediation of oil spills is a species of-

(A) Trichoderma (B) Xanthomonas (C) Bacillus (D) Pseudomonas

Open solution

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Q81. AIPMT 2006

Golden rice is a promising transgenic crop. When released for cultivation, it will help in

(A) Herbicide tolerance
(B) Alleviation of vitamin A deficiency
(C) Producing a petrol-like fuel from rice
(D) Pest resistance

Open solution

Q82. AIPMT 2005

Bacillus thuringiensis (Bt) strains have been used for designing novel -

(A) Bio-metallurgical techniques
(B) Biofertilizers
(C) Bioinsecticidal plants
(D) Bio-mineralization processes

Open solution

Q83. AIPMT 2005

Production of a human protein in bacteria by genetic engineering is possible because

(A) The human chromosome can replicate in bacterial cell
(B) The genetic code is universal
(C) Bacterial cell can carry out the RNA splicing reactions
(D) The mechanism of gene regulation is identical in humans and bacteria

Open solution

Q84. AIPMT 2005

Golden rice is a transgenic crop of the future with the following improved trait -

(A) High lysine (essential amino acid) content
(B) High protein content
(C) Insect resistance
(D) High vitamin – A content

Open solution

Q85. AIPMT 2004

DNA fingerprinting refers to :-

(A) Techniques used for molecular analysis of different specimens of DNA
(B) Anlysis of DNA samples using imprinting devices
(C) Molecular analysis of profiles of DNA samples
(D) Techniques used for identification of fingerprints of individuals

Open solution

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Q86. AIPMT 2004

The Ti plasmid is often used for making transgenic plants. This plasmid is found in

(A) Agrobacterium

(B) Rhizobium of the roots of leguminous plants

(C) Azotobacter

(D) Yeast as a 2 µm plasmid

Open solution

Q87. AIPMT 2004

ELISA is used to detect viruses, wherethe key reagent is :-

(A) Catalase (B) Southern

(C) DNA probes (D) Alkaline phosphatase

Open solution

Q88. AIPMT 2003

ELISA is used to detect viruses, where the key reagent is :-

(A) Alkaline phosphatase (B) Southern

(C) DNA probes (D) Catalase

Open solution

Q89. AIPMT 2002

Nucleus of a donor embryonal cell/somatic cell is transferred to an enucleated egg cell. Then after the formation of organism, what shall be true?

(A) Organism will have extra-nuclear genes of the donor cell

(B) Organism will have extra-nuclear genes of recipient cell

(C) Organism will have extra-nuclear genes of both donor and recipient cell

(D) Organism will have nuclear genes of recipient cell

Open solution

Q90. AIPMT 2002

Introduction of food plants developed by genetic engineering is not desirable because

(A) economy of developing countries may suffer

(B) these products are less tasty as compared to the already existing products

(C) this method is costly

(D) there is danger of introduction of viruses and toxins with introduced crop

Open solution

Q91. AIPMT 2000

The first successfully cloned mammals (animal) that gained worldwide publicity was

(A) Molly (a sheep) (B) Polly (a sheep) (C) Chance (a bull) (D) Dolly (a sheep)

Open solution

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Q92. AIPMT 2000

Producing a giant mouse in the laboratory was possible through

(A) gene mutation
(B) gene manipulation
(C) gene synthesis
(D) gene duplication

Open solution

Q93. AIPMT 1999

Hybridoma cells are :

(A) product of spore formation in bacteria
(B) hybrid cells resulting from myeloma cells
(C) nervous cells of frog
(D) only cells having oncogenes

Open solution

Q94. AIPMT 1999

The term 'humulin' is used for

(A) human insulin
(B) powerful antibiotic
(C) isoenzyme
(D) hydrolytic enzyme

Open solution

Q95. AIPMT 1999

The first transgenic crop was

(A) cotton
(B) pea
(C) tobacco
(D) flax

Open solution

Q96. AIPMT 1996

When scientists make an animal superior by view of genotype, introducing some foreign genes in it, is called

(A) immunization
(B) genetic engineering
(C) tissue culture
(D) biotechnology.

Open solution

Q97. AIPMT 1995

The transgenic animals are those which have

(A) foreign RNA in all its cells
(B) foreign DNA in some of its cells
(C) foreign DNA in all its cells
(D) both (a) and (b).

Open solution

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Answer Key

1. **B**	2. **A**	3. **A**	4. **C**	5. **D**
6. **A**	7. **C**	8. **C**	9. **C**	10. **C**
11. **C**	12. **B**	13. **A**	14. **A**	15. **C**
16. **A**	17. **B**	18. **D**	19. **C**	20. **B**
21. **D**	22. **B**	23. **C**	24. **D**	25. **D**
26. **C**	27. **C**	28. **B**	29. **D**	30. **C**
31. **B**	32. **C**	33. **B**	34. **A**	35. **D**
36. **D**	37. **C**	38. **C**	39. **C**	40. **D**
41. **C**	42. **B**	43. **C**	44. **B**	45. **B**
46. **C**	47. **B**	48. **B**	49. **A**	50. **D**
51. **D**	52. **B**	53. **C**	54. **D**	55. **B**
56. **C**	57. **D**	58. **D**	59. **B**	60. **B**
61. **C**	62. **A**	63. **A**	64. **A**	65. **A**
66. **B**	67. **C**	68. **B**	69. **D**	70. **D**
71. **A**	72. **D**	73. **C**	74. **D**	75. **C**
76. **D**	77. **A**	78. **A**	79. **D**	80. **D**
81. **B**	82. **C**	83. **B**	84. **D**	85. **C**
86. **A**	87. **D**	88. **A**	89. **B**	90. **D**
91. **D**	92. **B**	93. **B**	94. **A**	95. **C**
96. **B**	97. **C**			

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